Methods

Note: students must be prepared to answer examination questions focused on practical work, which will include questions about the practical exercises themselves and questions about the general methodological issues that are specified for this particular unit. This will include the following requirement.

a Identify, describe and apply

- i self-report
- ii scattergram
- iii correlation
- iv positive and negative correlation
- v procedure
- vi rating scales
- vii Spearman's test
- viii cross-sectional and longitudinal.

2.4 Biological approach

Students will be assessed on their ability to:

1 Definition of the approach	a Define the biological approach showing understanding that it is about the influence and impact of genes and the nervous system on individual differences.
	b Define and use psychological terminology appropriately and accurately including the terms: i central nervous system (CNS) ii synapse iii receptor iv neurone v neurotransmitter vi genes vii hormones viii brain lateralisation.

2 Methodology/ How Science Works

a Describe and evaluate twin and adoption studies as research methods.

b Describe PET and MRI scanning techniques.

c Identify, describe and apply the following terms:

i alternative, experimental and null hypothesis

ii one or two tailed with regard to tests

iii levels of significance (eg $p \leq 0.01, 0.05$)

iv Mann-Whitney U, - critical value and observed value

v dependent variable (DV) and independent variable (IV) in experiments

vi the use of control groups

vii experimental procedures including allocating groups to conditions (eg randomising) and sampling

vii levels of measurement.

Note: with regard to inferential tests, no calculations will have to be carried out in the examinations and formulae do not have to be learnt.

d Describe and evaluate, including strengths and weaknesses, the use of animals in laboratory experiments in the biological approach.

e Evaluate the use of animals in experiments in terms of credibility, ethical and practical issues.

f Evaluate the use of laboratory experiments in terms of validity, reliability and generalisability.

3 Content	a	Briefly describe the role of the central nervous system and neurotransmitters in human behaviour.
	b	Briefly describe the role of genes in behaviour (including the nature/nurture debate).
	c	With regard to gender development, describe the role of genes, hormones, and brain lateralisation.
	d	Evaluate the influence of biological factors on gender development including comparison with explanations from the Psychodynamic and Learning Approaches. Include the issues of use of animals and methodology in drawing conclusions.
4 Studies in detail	a	Describe and evaluate two studies relating to the Biological Approach. One must be Money J (1975) Ablatio penis: normal male infant sex-reassigned as a girl, and David Reimer's subsequent testimony and **one other**. This **must** be selected from the following: ■ Gottesmann I, and Shields J (1966) Schizophrenia in twins, 16 years' consecutive admissions to a psychiatric hospital ■ Raine et al (1997) Brain abnormalities in murderers indicated by positron emission tomography ■ Bellis et al (2001) Sex Differences in Brain Maturation during Childhood and Adolescence

5 Key issue	a	Describe **one** key issue of relevance to today's society and apply concepts, theories and/or research (as appropriate) from the Biological Approach to explain the issue. Suitable examples: ■ the debate over whether autism is an extreme male brain condition ■ the debate over whether transgender operations are ethical ■ the issue/debate of using drugs in pregnancy. **Note:** In the examination paper, students may be given stimulus material from a key issue to explain using concepts, theories and/or research (as appropriate) from the Biological Approach.
6 Evidence of practice: short practical	a	Devise and conduct one practical, which must be a test of difference collecting ordinal or interval/ratio data using an independent groups design. Suitable examples: ■ a comparison of test scores from males and females on verbal ability and spatial ability ■ comparing left and right handed people with their scores on tests of different abilities or preferences.
	b	Carry out a Mann-Whitney test and interpret the findings **Note:** with regard to inferential tests no calculations will have to be carried out in the examinations and formulae do not have to be learnt.
	c	Write up the hypothesis, results and analysis of the study using an appropriate graph and a table of the results. Draw brief conclusions, considering issues of validity, reliability, credibility and generalisability.

Methods

Note: students must be prepared to answer exam questions focused on practical work, which will include questions about the practical exercises themselves and questions about the general methodological issues that are specified for this particular unit. This will include the following requirement.

d Identify, describe and apply the following:

- i alternative, experimental and null hypothesis
- ii dependent variable (DV) and independent variable (IV)
- iii controls
- iv validity
- v reliability
- vi generalisability
- vii credibility
- viii levels of significance
- ix Mann-Whitney test
- x one and two tailed with regard to tests
- xi critical and observed values
- xii randomising and sampling
- xiii levels of measurement.